

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

How could activity distributed across cortex support one experienced situation? We separate three requirements: *coherence*, meaning collective phase order; *compatibility*, meaning that local descriptions agree where their domains overlap; and *self-representation*, meaning that the resulting state carries an accurate reconstruction of itself. The first supplies neither of the others, and four results establish that without any cortical hypothesis. Synchrony constrains content only through encoders shared on overlaps, and then by the resultant of the patch the bound is read on rather than of the whole population. The resultant does not determine the state: a winding field and a uniform one are stationary at one coupling with resultants spanning nearly the range of that observable. A contraction with a unique fixed point can discard its input, so accuracy across a family of states requires distinguishable codes. And where the encoding is a region's restriction, that resolution is demanded of the region before any readout is chosen. Under stated hypotheses, compatible local states then glue to a unique global state that an accurate contracting reconstruction fixes uniquely, and installed energy bounds the mean coupling a declared actuator supplies, so a coherent stationary branch requires it

to exceed $2D/\kappa$, with D the phase diffusion and κ set by the mode prices; energy is not the scarce side of that condition for a macroscopic substrate, which leaves it a statement about κ , derived here for neither cortex nor a processor. Lean 4 checks these results and their composition, with the eight connecting hypotheses as explicit arguments and no axioms beyond its libraries'. Further controls bound what observation can decide: distinct mechanisms share stationary phase distributions, and a coupling rule descending a squared-drift objective holds coherence while losing the structure it was meant to acquire. One prediction needs no absolute calibration: onset delay scales as the square root of the rate at which coupling crosses threshold, measured here as $\Delta K \propto v^{0.444}$, which a within-subject emergence-from-anaesthesia design can test. We propose the glued state and its internal reconstruction as correlates of unity and a minimal self, and an extracellular-field hypothesis linking awakening to rising coupling. Testing it requires calibrated geometry, resolved phase dynamics, validated decoders and a spatial observable beside the resultant.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their

domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter two relations as formal correlates of conscious unity and a minimal self; their interpretation requires evidence beyond the phase model.

Unity has been approached differently elsewhere: integrated information theory treats integration and exclusion as constitutive (Tononi et al., 2016), global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene et al., 2014), and predictive-processing accounts derive both from a hierarchical generative model (Friston, 2010). Here the three requirements are kept apart rather than derived from one another, so that each carries its own assumptions and admits its own measurement; Section 9.4 returns to the comparison.

Consider reaching for a cup. A visual description of its location and a description of the hand’s relation to it can agree about where the cup is while contributing different contents. Synchronizing the two processes would leave disagreement about that shared quantity unresolved. An internal representation must additionally locate the perceiver within the represented situation. This example motivates the requirements; the mathematical constructions use spatial measures and fields in real normed vector spaces, whose identification with perceptual content is a further modelling task.

The mathematical contribution is to specify connections between these requirements and the assumptions each connection needs. A declared actuator’s stored energy bounds its mean coupling. Phase encoders shared on

overlaps convert phase order into a bound on content disagreement. Compatible spatial measures have a unique global extension; approximate agreement controls a specified global selection. A contracting reconstruction makes that extension its unique fixed point only if it accurately reconstructs the extension. Accuracy across a family of states imposes a further condition on the information retained by the encoding.

The empirical contribution is to identify what observations can decide. Stationary phase distributions can be shared by stronger coupling, weaker diffusion and a matched common drive. Finite ramps change the observable onset. A winding field is stationary at the same coupling as a uniform one and has the opposite resultant, so that observable does not determine the state. Coherent adaptation can lose alignment with a designated environmental template, and a compatibility estimator needs diagnostics that distinguish content from decoder artifacts. These findings determine the measurements needed to test the proposed connections.

Three of the ingredients are standard: the synchronization threshold $K_c = 2D$ for the noisy mean-field model (Sakaguchi, 1988), the unique gluing of compatible local sections, and the Banach fixed-point theorem (Banach, 1922). Four things here are not. The installed-energy condition bounds mean coupling by the stored energy of a declared actuator, which makes a resource condition necessary for coherence on the stationary branch. The coherence-to-content bound converts phase order into a statement about descriptions rather than about timing, and its patch-local form is what keeps

58 it from being vacuous in a large population. A stability criterion on the
59 reweighted coupling kernel, together with an existence condition on an am-
60 plitude layer, settles when a low resultant reports weak coupling and when
61 it reports a winding field. And the composition is machine-checked together
62 with a witness discharging all eight hypotheses at once, which is what makes
63 their conjunction non-empty rather than a list no object satisfies.

64 Granting those premises, collected in Table 1, the account runs as follows.
65 Extracellular contraction on awakening raises endogenous field amplitude
66 and with it the effective coupling between cortical oscillators. Once that
67 coupling passes $2D$, a coherent phase branch exists, and from an initial spread
68 within a quarter turn the population settles onto it; other initial data leave
69 windings stationary at the same coupling. Encoders shared where cortical
70 territories overlap turn that phase order into agreement about the quantities
71 the territories describe in common, so the local descriptions glue: there is
72 one state of which each is a restriction. Part of that state encodes the state
73 itself, and a readout of that part reconstructs it accurately enough, and
74 sensitively enough across the states that matter, to make it the unique state
75 the system models itself as being. On this account the glued state is what
76 makes distributed activity one situation, and its accurate self-reconstruction
77 is what gives that situation a perspective. Every step of it is conditional.

78 Section 2 collects the results that hold whatever physical field is proposed
79 and whatever substrate carries it; a reader who rejects the cortical hypothesis
80 of Section 8 still has to accommodate them. Sections 3–5 then develop the

81 three requirements together (Fig. 1). Section 6 composes them, stating the
 82 thermodynamic constraints and all eight connecting hypotheses per branch;
 83 its supporting models of feedback, learning and resource supply have distinct
 84 state spaces and assumptions. Section 7 then examines what the observables
 85 can establish, before Section 8 develops the cortical hypothesis and an awak-
 86 ening experiment. The hypothesis is that extracellular contraction increases
 87 field-mediated coupling; its direction, rate and relation to recovery require
 88 independent calibration. Supplemental Material organizes the constructions
 89 and controls behind the claims they support.

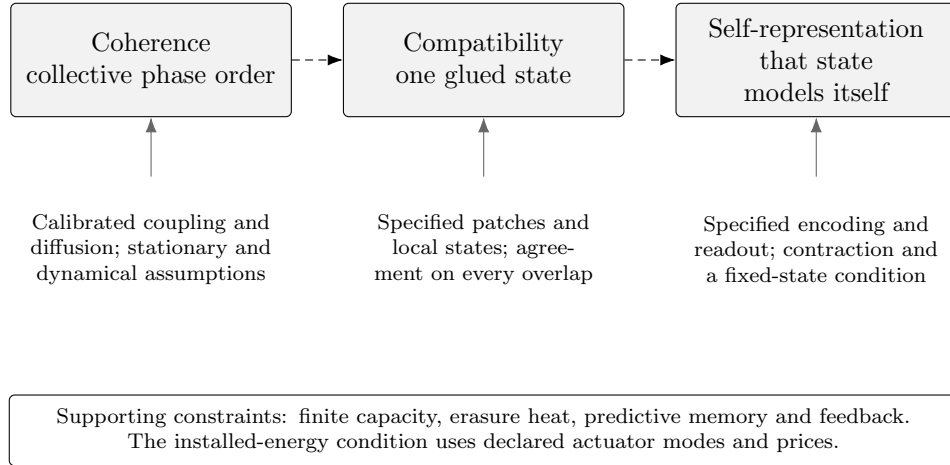


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

90 2 What holds without the cortical hypothesis

91 Most of what follows is conditional. Table 1 names eight hypotheses that
92 the composition consumes rather than derives, and the cortical identification
93 of Section 8 is a further commitment on top of them. Five results are not
94 conditional in that way. They constrain any account that reads unity off
95 phase coherence, whatever physical field is proposed to carry the coupling
96 and whatever substrate carries the field. Each is stated below where it is
97 used; collected here, they are the article’s fixed points.

98 **Phase order alone does not constrain content.** Assign every patch
99 the same phase and let two local mass profiles disagree at a shared site.
100 The profiles are chosen independently of phase, so synchrony bounds their
101 disagreement not at all, and they have no common extension: perfect locking
102 is consistent with local descriptions that do not glue (Section 4). What the
103 positive result of Section 4.2 costs is exactly this gap. A bound on content
104 disagreement needs encoders already related on overlaps, and that relation
105 is a supplied fact about the descriptions rather than a consequence of the
106 dynamics.

107 **The order parameter does not determine the state.** A field winding
108 once across a periodic sheet has $r = 0$ while every patch of it stays locked,
109 and this is not a contrived initial condition. On a periodic array of any
110 dimension, a winding is stationary for *any* isotropic symmetric kernel at

111 identical frequencies, because the coupling term is odd in the separation
 112 against a kernel even in it (`char_is_kuramoto_trajectory`). Section 7.3
 113 exhibits a sheet on which uniform and winding initial conditions reach 0.9657
 114 and 0.0083 at one coupling, diffusion and kernel. A low resultant therefore
 115 reports either weak coupling or a winding field, and no measurement in r
 116 alone separates them.

117 **A unique fixed point is not a self-model.** A constant readout con-
 118 tracts, fixes whatever state it returns, and reconstructs nothing. Uniqueness
 119 is consequently the wrong criterion, and Section 5.2 gives the right shape
 120 of one: reconstructing M states pairwise more than 2ε apart within error ε
 121 requires M distinguishable codes, so an alphabet of 2^b codes caps the family
 122 at $M \leq 2^b$. This is a counting argument and holds for any encoder and
 123 readout whatever.

124 **Where the encoding is a restriction, the region must resolve the**
 125 **family.** Read the code as the state's own restriction to a region and
 126 the criterion moves off the readout entirely. A boundary whose region
 127 reads two relevant states identically cannot reconstruct them more than 2ε
 128 apart (`not_isRestrictionResonance_of_reconstructs`), and a resonant
 129 boundary that does reconstruct a separated family restricts injectively on
 130 it (`restrictToAvatar_injOn_of_isRestrictionResonance`). Where the
 131 region sits is fixed before an encoding is chosen, so no readout supplies
 132 resolution the region lacks.

133 **Agreement after a change takes communication.** In a synchronous
134 network whose updates read only the site’s own state and its declared neigh-
135 bours’ messages, two configurations agreeing on a site’s causal past through
136 T rounds leave that site identical at round T . Two interventions outside
137 that past that require reports more than 2ε apart therefore cannot both be
138 reported from it within ε by the deadline (Section 5). Snapshot compatibil-
139 ity and timely compatibility are therefore different requirements, and cutting
140 the path removes the second while leaving the first intact.

141 None of these depends on the coupling being electromagnetic, on the
142 substrate being cortex, or on the eight hypotheses of Table 1 holding. Sec-
143 tion 9.3 applies the last three to a candidate that is neither biological nor
144 field-coupled.

145 **3 Coherence: physical coupling and its re-** 146 **source condition**

147 **3.1 Dynamics and the stationary threshold**

148 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

149 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and
 150 K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field
 151 coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct
 152 parameters. The stationary calculation below uses identical frequencies and
 153 this scalar coupling. Applying it to a spatial kernel requires an aggregation
 154 rule and a check of the resulting approximation.

155 For identical frequencies in their rotating frame, the positive normalized
 156 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

157 where I_n are modified Bessel functions and ψ is the collective phase. The for-
 158 mal derivation uses the absence of stationary current for a periodic gradient
 159 drift to obtain this density, then verifies normalization and self-consistency.
 160 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988). Above
 161 threshold, exactly one positive solution coexists with zero and increases
 162 strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

163 Figure 2 shows this stationary branch.

164 Linearization about the uniform density gives rate $K/2 - D$ for the first
 165 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
 166 exactly above threshold. About the coherent density, the quadratic form

167 is strictly dissipative transverse to collective rotation, whose rate is zero.
168 These linear estimates concern the limiting equation. Relating them to finite
169 stochastic trajectories requires a dynamical mean-field limit and branch selec-
170 tion; exchange symmetry alone does not establish the needed independence
171 (Sznitman, 1991).

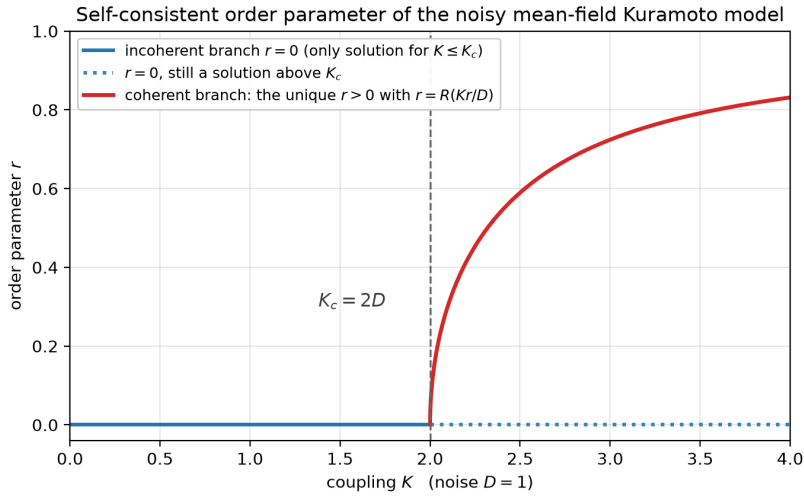


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

172 3.2 Installed energy bounds mean coupling

173 A spatial model replaces the finite sum by integration against a kernel. One
174 declared implementation carries response modes with continuous profiles ϕ_i ,

175 installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

176 The same configuration sets the reciprocal kernel and stored energy. An
 177 executed transition changes the kernel through its changed occupancies and
 178 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The
 179 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-
 180 pair reconstruction on refining spatial partitions converges to its product-
 181 measure integral under the specified compactness and sampling assumptions;
 182 the refinement index is independent of the number of control updates.

183 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 184 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 185 integral gives

$$\begin{aligned} U_{\text{inst}} &= \sum_z p(z) U(z), \\ K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}. \end{aligned} \quad (5)$$

186 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 187 coherent stationary branch and any growing Fourier mode about the uniform
 188 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 189 is necessary: energy above it need not produce sufficient coupling because
 190 the bound can be loose.

191 The quantity constrained is installed energy rather than dissipated heat,
 192 and how much the condition excludes depends entirely on κ , which the mode
 193 profiles, prices and substrate normalization supply. Any macroscopic sub-
 194 strate carries installed energy far above the thermal floor, so the inequality
 195 binds through that conversion and not through the energy, and no result here
 196 derives κ for cortex or for any other candidate; what the condition contributes
 197 is a resource requirement in units a measurement can reach. Applying the
 198 scalar threshold to heterogeneous or signed spatial dynamics requires a jus-
 199 tified reduction, and identifying the installed kernel with a cortical field is a
 200 separate physical commitment.

201 **4 Compatibility: from coordinated activity** 202 **to a global description**

203 **4.1 Why common timing is insufficient**

204 For overlapping patches U and V , compatibility means that restricting either
 205 local description to $U \cap V$ gives the same object. Descriptions can agree on
 206 shared quantities while contributing different information elsewhere. Com-
 207 patible sections of the specified sheaf have a unique global extension. On
 208 a finite cover, local finite spatial measures agreeing as measures on every
 209 overlap also have a unique finite-measure extension whose sheaf image is the
 210 glued section. No mass-one normalization is imposed.

211 In the cup example, visual and reaching descriptions must express a
 212 shared quantity, such as cup location, in common coordinates. Restriction
 213 retains that quantity while dropping details particular to each description.
 214 Agreement permits surface appearance and hand configuration to contribute
 215 to one situation. The mathematical counterexample assigns the same phase
 216 to every patch while two local mass profiles disagree on a shared site. Those
 217 profiles have no common extension. Because they are chosen independently
 218 of phase, synchrony places no constraint on their disagreement.

219 This distinction limits the biological interpretation. Receptive territories
 220 and content domains must be related to spatial patches, and the descriptions
 221 must support the required extension. Matching overlap marginals of de-
 222 coded probability laws need not determine a unique global joint distribution
 223 (Abramsky & Brandenburger, 2011).

224 **4.2 When coherence does constrain content**

225 A declared relation between phase and content gives a positive result. Let
 226 contents take values in a real normed vector space E . Patch i reads an
 227 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 228 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 229 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 230 sites; their equality on overlaps is a supplied relation between descriptions of
 231 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 232 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every

233 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1-r^2}. \quad (6)$$

234 Here discrepancy is the norm of the difference, and r and N are read on
235 whichever population the bound is applied to. That choice is what decides
236 whether the estimate says anything. A patch carries an order parameter of
237 its own, and reading the bound there carries that patch's resultant and that
238 patch's site count (`chord_le_of_patch_coherence`, `compatible_of_patch_`
239 `coherence`), so a large population observed through small patches is con-
240 strained by the small count. Read instead on the whole array, the factor N
241 makes the estimate weak, because it draws a pointwise guarantee from an
242 average over N^2 pairs. At $r = 1$ the observations agree exactly at either
243 scale. A shared scalar encoder is a special case. Dropping the overlap rela-
244 tion permits disagreement even at perfect locking. A common nonexpansive
245 linear projection preserves agreement, but agreement of a projection does
246 not establish agreement of the full vector.

247 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
248 constrains anything only where $\sqrt{2} N\sqrt{1-r^2}$ falls below that. The winding
249 sweep of Section 7.3 fixes both ends of that range on one sheet. Read on the
250 sweep's own patch, 625 sites at patch-local order 0.9015, the bound is $382.5 L$
251 and constrains nothing; holding that order fixed, the count alone brings it
252 below the maximum chord only at 3 sites or fewer. A nearest-neighbour pair

253 of the same winding, the smallest patch there is, gives $0.1388 L$. Shrinking
 254 patches therefore thins the overlaps at the same rate as it tightens the bound,
 255 and the two ends are not reached by the same cover. At a columnar grain
 256 the estimate constrains agreement inside one locally synchronous population
 257 and is vacuous across the distant territories whose agreement unity is about.

258 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 259 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 260 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 261 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

262 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 263 duce an initial mismatch. Approximate observations leave a nondecaying
 264 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 265 update to the phase process.

266 4.3 What approximate agreement determines

267 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 268 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 269 Weights that vanish outside their patches and sum to one at each site select
 270 a global profile by weighted averaging. The selection lies within ε of each
 271 patch there, and any two such selections differ by at most ε , independently
 272 of cover multiplicity. An L -Lipschitz readout consequently varies by at most

273 $L\varepsilon$ across selections.

274 At zero tolerance, the selection is the unique exact extension. At every
275 positive tolerance there are families with no exact extension; selection then
276 gives an approximation whose dependence on weights is bounded. Supple-
277 mental Material gives a three-site witness attaining the bound. The mass-
278 profile construction uses the scalar case of the content dynamics and addi-
279 tionally requires nonnegativity of the initial profiles and observations. The
280 general vector result supplies no identification with the spatial-measure sheaf.

281 These constructions concern declared objects: a specified cover, contents
282 valued in a normed space, and encoders related on overlaps. They select
283 neither the neural cover nor the decoder. Identifying their contents with
284 perceptual content, and establishing that the identification bears on experi-
285 ence, is a further modelling task; Section 7 sets out what a measurement of
286 it would have to establish.

287 **5 Self-representation: a state that models it-** 288 **self**

289 **5.1 The reflexive construction**

290 Compatibility gives a global state. Reflexivity relates it to an encoding within
291 that state. Equip global sections with a complete metric. If the additional
292 restriction-resonance condition identifies encoding with restriction to a region

293 U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \quad (8)$$

294 The region and readout are fixed across the state space. If F contracts and
295 fixes the section s_* formed from the local descriptions, Banach's theorem
296 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
297 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
298 Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
299 one for $\tau > 0$. Constructing a map with that factor remains a modelling
300 obligation.

301 In the cup example, the global state includes the represented scene and the
302 perceiver's relation to it. A local encoding must retain enough information
303 for the readout to recover that modelled state; cup position alone would be
304 insufficient if hand configuration and viewpoint could not be reconstructed.

305 For the same map with Lipschitz factor $q < 1$, approximate reconstruction
306 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (9)$$

307 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
308 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
309 weakens.

310 5.2 Reconstruction across a family of states

311 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 312 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 313 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

314 Thus M relevant states pairwise more than 2ε apart require at least M
 315 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 316 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 317 existing self-prediction map.

318 This is an additional accuracy criterion on a family, with no requirement
 319 for multiple exact fixed points. A blind encoder or constant readout fails it
 320 for separated states despite having a unique fixed point.

321 Where the encoding is read as the state's own restriction to a region, the
 322 criterion becomes a condition on the region. Restriction resonance requires
 323 the encoder and that restriction to identify the same pairs of states, so a
 324 boundary reconstructing two relevant states more than 2ε apart while its re-
 325 gion reads them identically is not resonant (`not_isRestrictionResonance_`
 326 `of_reconstructs`); read forwards, a resonant boundary that reconstructs
 327 a pairwise separated family restricts injectively on it (`restrictToAvatar_`
 328 `injOn_of_isRestrictionResonance`). Which states a region reads identi-
 329 cally is fixed by where the region sits, before an encoding is chosen, so the

330 resolution the family demands is the region's and no readout supplies it. The
 331 implication runs one way: a region that does resolve the family is not thereby
 332 shown to carry a resonant encoding, and E89 supplies no such resolution.

333 **5.3 Agreement after a change takes communication**

334 Snapshot compatibility does not specify how quickly descriptions respond
 335 when a shared quantity changes. In a synchronous network whose updates
 336 read only the site's own state and declared neighbours' messages, two initial
 337 configurations agreeing on a site's causal past through T rounds produce the
 338 same state there at round T . If two fresh interventions outside that past
 339 require reports separated by more than 2ε , no report from that site can be
 340 within ε of both. This follows from the reconstruction bound in Section 5.2.

341 A three-site delay line witnesses failure after one round and exact report-
 342 ing after two; cutting the path prevents delivery at every round. A report
 343 may still happen to be correct about one fixed world. The bound concerns
 344 responsiveness across interventions.

345 Throughout this section *prediction* names a reconstruction map; reading
 346 it as temporal neural prediction would require dynamics. The relevant family
 347 of states, the metric, the tolerance, the communication graph and the dead-
 348 line are all declared inputs, as are the represented variables and the fidelity
 349 counted as accurate. Reading the resulting code count as physical memory
 350 capacity requires a hardware model, and a readout that varies with the state
 351 requires a further one.

352 6 Thermodynamics and composition

353 The phase and representation results concern coupling, content disagreement
354 and reconstruction error. Thermodynamics constrains the physical opera-
355 tions that could implement them, and it does so from the side rather than
356 along the spine: no edge below derives coherence, compatibility or reconstruc-
357 tion from a heat bound, and the resource condition these edges terminate in
358 is necessary without being restrictive for any macroscopic candidate (Sec-
359 tion 3.2). What this half contributes is a ledger in which a resource claim
360 is stated in checkable units. Finite-register capacity obeys $H(p) \leq \log |X|$;
361 erasure heat depends on specified system–bath assumptions (Landauer, 1961;
362 Norton, 2005; Shenker, 2000). A separate boundary result concerns a field
363 reaching disconnected vacuum components (Kibble, 1976). The composition
364 connects these objects through the explicit hypotheses in Table 1.

365 6.1 Feedback changes the information constraint

366 For passive memory X_t driven by an environmental signal S_t , the predictive
367 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

368 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
369 et al., 2012). Zero retained information also satisfies it, so a task and a
370 learning rule require additional specification. Relating this bound to Lan-
371 dauer heat requires a common thermal scale and budget, and memory waste

372 bounded by erased entropy.

373 An acting agent changes the signal it will receive. Specify policy, world,
374 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
 I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
 &\quad - I(X_t; S_{t+1} \mid S_t).
 \end{aligned}
 \tag{12}$$

375 The second conditional term vanishes in the passive model. With feedback,
376 the signed difference can be negative as action creates correlations with the
377 future. A causal reading requires an environmental state sufficient to describe
378 the relevant dynamics.

379 For a controlled substep, the active thermodynamic bound instead uses
380 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
381 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
382 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
383 2013). Every forward path of positive probability must have a positive-
384 probability reverse for this finite identity. A missing reverse gives infinite
385 divergence. The supplement develops the support conditions and composes
386 actuation with memory updates on their shared intermediate law.

387 **6.2 Learning and repeated operation need explicit re-** 388 **sources**

389 The finite examples distinguish selecting a feasible policy, relaxing a register
390 toward an encoded reward, and learning an unknown task from observa-
391 tions. The observational learner acquires correlation with an environmental
392 parameter while its action marginal stays uniform. A continuing model exe-
393 cutes actuation, observation-driven learning and reset in sequence, preserving
394 above-chance performance at a positive work cost per cycle.

395 A phase-derived sensor makes the informational distinction concrete: an
396 informative readout and a blind control have the same phase order but dif-
397 ferent learning performance. Attaching the learner’s register to the actuator
398 of Section 3.2 ties reward, installed kernel and work to its own executed law.
399 The learning witness rises above the necessary installed-energy bound while
400 its matched control remains on the exclusion line. These finite constructions,
401 detailed in Supplemental Material, leave the evolving feedback between phase
402 dynamics, installation and sensing to be specified.

403 For that model, common energy E , thermal scale θ , local detailed balance
404 and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (13)$$

405 where expected cumulative work stays within the supplied allowance B at
406 every prefix. This bounds expectations. Separate constructions track a store

on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles, while continuing operation requires replenishment. Register–bath witnesses also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 3.2 requires physical relations that these separate models do not supply automatically.

6.3 What the composition establishes

The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named substep at the register’s thermal scale satisfies

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

with B the register’s heat. The upper allocation follows from a ledger whose operations deliver that total heat to one reservoir and whose other operations do not increase joint entropy. Those ledger assumptions are physical inputs. Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness.
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Refinement and physical identifications remain inputs.
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements.

Table 1 (continued)

Connection	Status	Required assumption
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and readout. They conclude **UnifiedSelf**: the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate resonance condition of Eq. (8).

Each of the eight edges is supplied rather than derived from the one above it. E12 is equivalent to its conclusion, so capacity and boundary are independent roots rather than a step from one to the other; both E45 discharges take their convergence from refinement data and not from the thermodynamic bound; E78 states that coherence relaxes the supplied cover, and the coherence-to-content estimate of Eq. (6) bounds disagreement only for encoders already shared on overlaps. The inferential content of the composition is its last step, where contraction and a fixed glued section give uniqueness. What composing them establishes is that the premises are consistent, named

440 in one place and machine-checked together, and not that coherence follows
441 from installed energy or self-representation from coherence.

442 Joint witnesses combine a register, boundary, refining mesh and small
443 spatial cover; active witnesses add controlled thermal updates and a priced
444 installation. They establish satisfiability under their stated identifications.
445 Mesh convergence follows from refinement data, and the witnesses establish
446 no common cortical mechanism. Supplemental Material maps the claims to
447 their Lean declarations and records the counterexamples. The development
448 declares no additional axioms beyond its libraries' foundational axioms.

449 **7 What the observations establish**

450 The composition holds under its premises. Whether observation can estab-
451 lish those premises is a separate question, and the three requirements it
452 connects concern different objects: phase dynamics, local descriptions and
453 internal encodings. Their observables must retain these distinctions. The
454 following controls show which mechanisms remain indistinguishable, which
455 temporal approximations fail, which states one observable cannot separate,
456 and what independent content measurements must establish. Each is a finite
457 numerical regime on a declared model, with full parameters and controls in
458 Supplemental Material, and none of them calibrates a cortical quantity; the
459 scope notes below concern what each result does and does not separate.

460 7.1 Stationary distributions leave the mechanism un- 461 determined

462 Equation (2) gives a concentration–coherence relation with no fitted shape
463 parameter. Estimate r from circular resultant length and a from density
464 shape: inverting R from the measured r would make agreement automatic.
465 Even independently estimated agreement identifies a statistical family shared
466 by different mechanisms.

467 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
468 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
469 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
470 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
471 and diffusion D_0 have that same density. Their stationary distributions co-
472 incide window by window despite different causes.

473 Resolved dynamics supply additional information. In the ideal model with
474 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
475 increment means constrain drift. Separating endogenous coupling from a
476 matched common drive additionally requires a measured drive or controlled
477 perturbation; identifying electromagnetic coupling requires evidence specific
478 to that interaction. Supplemental Material gives the equivalence construction
479 and its numerical check.

480 **7.2 Finite ramps change the observable onset**

481 The square-root law in Eq. (3) describes stationary solutions. A changing
482 coupling requires time for the phase distribution to approach those solutions.
483 In a deterministic mean-field control that crosses the threshold symmetrically,
484 none of the scanned rates meets the declared tracking tolerance: slowing
485 the crossing does not remove the terminal residual under that initialization.
486 A slow control variable therefore supplies no general guarantee of branch
487 tracking at onset. Supplemental Material gives the residual, rate range and
488 admissible rates away from threshold.

489 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
490 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
491 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
492 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The
493 6 uncensored legs supply the delay scaling below; the decade-spaced ones
494 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
495 onset exponents differ from those of the exact stationary branch evaluated
496 on the same grids. At the fastest speed the ensemble mean never reaches the
497 fit window, so its fitted slope does not estimate stationary onset.

498 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
499 makes the apparent exponent depend on ramp speed and fitting window.
500 The specified deviation-based critical ramp speed is not bracketed by this
501 sweep. With fluctuating coupling, the relevant stationary comparison also
502 depends on rate: the mean coupling approximates order in the tested fast

503 regime, while averaging stationary order over coupling works for the tested
 504 slow drives confined to one side of threshold. Slow crossing drives can satisfy
 505 neither approximation.

506 **The prediction.** The delay scaling is what this section supports as a pre-
 507 diction, and it needs no absolute calibration. Over the 6 uncensored legs the
 508 mean delay follows $\Delta K \propto v^{0.444}$, with a 95% slope interval of $[0.394, 0.494]$.
 509 The reference for that exponent is the same estimator applied to a deter-
 510 ministic mean-field run seeded at the fluctuation floor, which returns 0.425
 511 over the same legs at the same escape criterion, inside the interval on the
 512 measurement. The distance from $1/2$ is what an absolute escape criterion
 513 costs: the run crosses the saturation between the floor it starts from and the
 514 level it is stopped at, and tightening the criterion to $r \geq 0.05$ returns 0.490.
 515 Stated as a law rather than as a number: onset delay scales as the square
 516 root of the rate at which coupling crosses threshold. Testing it requires only
 517 that $\dot{K}$ be monotonic in a manipulated rate — not its value, nor κ , nor γ
 518 — so it is a within-subject design that varies the rate of emergence from
 519 anaesthesia alone, with each participant as their own control. Such a design
 520 must avoid the hazard this sweep exhibits: a fixed absolute escape criterion
 521 meets a size-dependent critical fluctuation floor, so the criterion has to be
 522 referred to each recording’s own pre-threshold fluctuation scale rather than
 523 to a shared number. The diffusion D is the remaining input, and Section 7.1
 524 notes that it is estimable from resolved phase traces through the rate $2D$

525 at which quadratic variation accumulates, which is a different quantity from
 526 the concentration the propofol exercise could not range over.

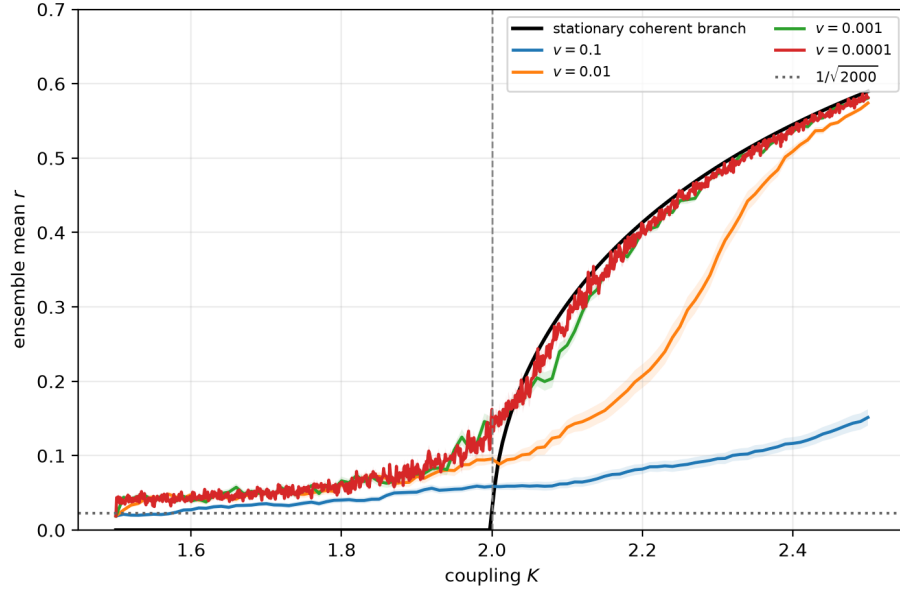


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

527 Additional controls address integration bias, population size and spatial
 528 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 529 $[1.958, 2.009]$ contains 2.

530 **7.3 A winding state leaves the order parameter am-** 531 **biguous**

532 The order parameter r is indifferent to where a phase sits in space. A field
533 winding once across a periodic sheet therefore has $r = 0$ while every patch
534 of it stays locked. Such a field is not an artificial construction here. On a
535 torus a winding is stationary for any isotropic symmetric kernel at identical
536 frequencies: the coupling term is odd in the separation against a kernel even
537 in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves
538 this for any finite periodic array, this sheet included. A winding satisfies the
539 positivity and identical-frequency hypotheses. It violates only the quarter-
540 turn initial spread that the convergence result additionally requires before
541 placing the limit at the uniform state.

542 On the sheet of Section 3's spatial control, a winding survives below
543 an interpolated decay length of 0.1871 mm and unwinds above it. That
544 boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same sheet at the
545 same coupling, diffusion and kernel reaches $r = 0.9657$ from a uniform start
546 and $r = 0.0083$ from a winding one. Both are stationary, with patch-local
547 order 0.9015 and no winding defect. At fixed K the observable therefore takes
548 two values spanning nearly its whole range. Coupling does not determine the
549 resultant, and a low resultant does not by itself distinguish weak coupling
550 from a winding field.

551 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-

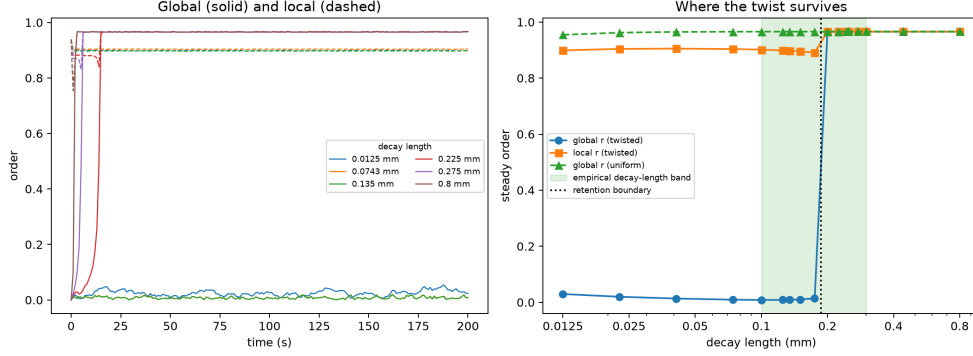


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

552 trol carries, the boundary moves to 0.1421 mm. The same comparison at
 553 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.
 554 Heterogeneity shortens the range that sustains a wave without removing it.

555 Which windings persist has a criterion behind it. Linearizing the phase
 556 model about a winding returns a coupling of the same translation-invariant
 557 form, each separation reweighted by the cosine of the phase the winding
 558 advances across it (`hasDerivAt_charJacobian`). That reweighted kernel is
 559 again even in the separation, so the same characters diagonalize it and ev-
 560 ery mode’s rate is one of its eigenvalues. Where the winding advances by
 561 at most a quarter turn across every separation the coupling reaches, the

562 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`
 563 `stable_of_quarter_turn`); where some rate is positive, that mode grows
 564 without bound (`charJacobian_growing_mode`). Moving coupling mass to
 565 longer separations is what carries a phase advance past a quarter turn, and
 566 that is the mechanism the decay-length sweep varies. The criterion bounds
 567 rates rather than trajectories: asymptotic stability is not proved, so the
 568 states that persist occupy a narrower range than it admits, and the bound-
 569 ary quoted above is measured and not derived from it.

570 The same reweighted kernel measures the potential's curvature. A wind-
 571 ing has $r = 0$ where a global minimizer of the Lyapunov potential has $r = 1$,
 572 so it is not one (`char_not_potential_min`): the quarter-turn condition on ini-
 573 tial data does not merely decline to cover a winding, it must exclude it. Along
 574 a mode, the potential's second variation is that kernel's quadratic form, so a
 575 growing mode is a direction of falling potential and a winding carrying one
 576 is not a local minimum either (`char_not_local_min_of_charLambda_pos`).

577 An amplitude layer decides which of these states exist, and leaves what
 578 the resultant reports unchanged. A phase state carries no amplitude. Ev-
 579 ery oscillator sits on the unit circle, so a field with an amplitude node has
 580 no representation in these dynamics. The real Stuart–Landau extension
 581 supplies one on the same coupling, and leaves intact the gradient struc-
 582 ture the convergence results rest on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i =$
 583 $(\mu - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$. Its phase equation is the Kuramoto equation
 584 with each coupling weighted by a ratio of amplitudes (`amplitude_phase_is_`

585 kuramoto). The phase model is this one with the amplitude ratios set to
 586 one, rather than a separate object.

587 What the added degree of freedom decides is which windings exist at all.
 588 On a periodic array of any dimension, this sheet included, a winding requires
 589 $\mu + \lambda > 0$. Here $\lambda = \sum_d f_d(\cos \phi_d - 1)$ is the eigenvalue the kernel supplies,
 590 summed over separations, with ϕ_d the phase the winding advances across d
 591 (exists_charState_iff; on the ring $\phi_d = 2\pi qd/n$). For the nonnegative
 592 kernels used here $\lambda \leq 0$, and moving coupling mass to longer separations
 593 lowers it further. A broad enough kernel therefore removes the winding out-
 594 right. The sweep above reaches that same boundary by a different route:
 595 the phase model retains every winding at every coupling, and loses this one
 596 under integration instead. The two layers bound the same regime from op-
 597 posite sides. A phase recording does not separate a state the amplitude layer
 598 excludes from one the dynamics fails to sustain.

599 These states are windings and not spirals, and on the sheet that is a
 600 theorem rather than a shortage of dimensions. A cortical phase singularity
 601 is a point at which the amplitude vanishes and around which the phase turns
 602 once, read in practice as the circulation around a small plaquette. The states
 603 above have equal modulus at every site, so that circulation is exactly zero
 604 on every plaquette (plaquetteCirculation_eq_zero_of_winding). Nor can
 605 the layer host a single defect. Over a periodic sheet the circulations of *any*
 606 phase field sum to zero, so defects arrive in cancelling pairs (no_isolated_
 607 defect). The sweep's near-zero measured defect density agrees with the first

608 of these and says nothing beyond it.

609 Two limits bound the layer for a reader of the cortical literature. Relax-
610 ational dynamics has no propagation speed, which is the quantity a cortical
611 wave measurement leads with. And $|z|$ is a normal-form radius, which a mea-
612 sured envelope is not; this work identifies the two nowhere. Supplemental
613 Material gives the sweep, its uniform control, the amplitude layer and the
614 standing wave it admits inside the band, together with the remaining limits
615 of both.

616 A vanishing resultant is the absence of global agreement and not of agree-
617 ment. Eq. (6) read on a nearest-neighbour patch of a winding gives a dis-
618 crepancy bound of $2\sqrt{2}L|\sin\phi|$, with ϕ the phase advanced across the patch,
619 which falls to zero as the same winding is spread over more sites (`chord_le-`
620 `of_char_patch`), while the global form of the same bound is vacuous at $r = 0$.
621 Encoders shared on overlaps therefore still bound content disagreement in-
622 side a patch of a state that no global observable calls coherent. The winding
623 number itself stays unconnected to content.

624 This qualifies what a phase trace can establish. An onset criterion stated
625 in r alone registers a change of basin as readily as a change of coupling.
626 The protocol of Section 8.3 therefore needs a spatial observable beside the
627 resultant.

628 **7.4 Coherent adaptation can lose environmental align-** 629 **ment**

630 An adaptive coupling rule might supply representational structure, but its
631 objective must reward the relevant agreement. For symmetric coupling, pair-
632 wise sine terms cancel in the population mean drift. Minimizing squared drift
633 at that fixed mean therefore rewards equalizing drift across nodes. In the
634 frequency-cluster control, this favors coupling between unlike-frequency clus-
635 ters, whereas the diagnostic environmental template rewards coupling within
636 clusters.

637 The formalization proves descent of the squared-drift objective for a spec-
638 ified gradient flow at fixed phases. Its functional differs from thermodynamic
639 entropy production: at a coherent stationary density the probability current
640 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
641 rial gives the operator, its gradient and the conditions of the descent result.

642 The finite control co-evolves noisy phases and symmetric, nonnegative
643 couplings at fixed total resource. Its template is a diagnostic covariance of
644 latent signals, never a training input. Across three seeds, second-half order
645 remains between 0.9167 and 0.9222, while the within-cluster to between-
646 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 5).
647 Random-update, permuted-gradient and frozen-kernel controls are reported
648 in Supplemental Material.

649 The immediate post-update objective reduction disappears when aver-

aged over complete update intervals. A prespecified grid of 6 rate-cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

7.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on constructed data separates these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial; the compatible construction scores 0.476 on an interval containing one half. Shared-prior shrinkage can nevertheless manufacture agreement, and regional bias can manufacture disagreement. Per-patch decoding diagnostics detect these failures; the compatibility statistic alone does not.

Phase agreement cannot replace that measurement. On constructed con-

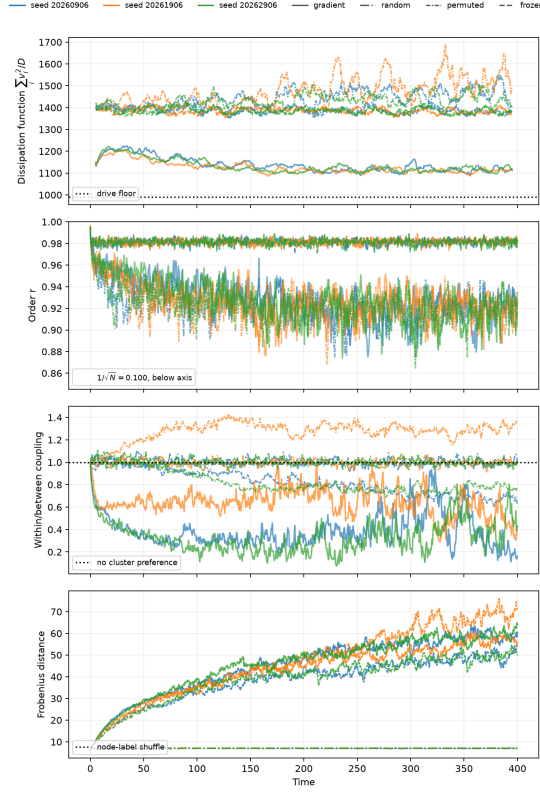


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

672 tent, phase carries a share 1.000 of the content entropy when content is built
673 from it and 0.000 when it is not. A phase-derived statistic reports com-
674 patibility on phase-locked patches whose local descriptions disagree. The
675 implemented content statistic detects their incompatibility. These controls
676 validate a measurement on known constructions; a neural content model and
677 decoder remain to be established.

678 Choosing the cover also chooses a candidate subject: trivial descriptions
679 can agree while carrying little content, and a partition can conceal disagree-
680 ment. The useful choice must therefore be constrained by what neural ac-
681 tivity encodes. The exact composition assumes compatibility of its supplied
682 cover. The content mechanisms above identify additional hypotheses under
683 which agreement can arise; they leave the empirical choice of content and
684 the connection to that cover explicit.

685 **7.6 What an informative phase-density measurement** 686 **requires**

687 The concentration range and the dependence structure decide whether the
688 concentration–coherence relation can be tested at all. Under the specified
689 log-density estimator and bias-floor criterion, discrimination at the observed
690 concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned
691 designs, rising to 31000 when sites share a mean in groups of 10. These
692 counts describe that estimator and dependence model; the intended recording

693 requires its own calibration. Supplemental Material reports the bin-count
694 controls and the full dependence-aware design.

695 That ceiling comes from exercising the estimators on public propofol-
696 sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms windows.
697 Bipolar and common-reference montages produce different residuals. The
698 bipolar results are compatible with the Bessel relation over a concentration
699 range too narrow to distinguish it from its linear approximation at the ob-
700 served scatter, which is the range the design scan takes as given. Temporal
701 pooling changes the estimand, and sedation recordings leave the awakening
702 hypothesis untested.

703 An informative dataset therefore needs sufficient concentration range or
704 precision, calibrated estimators and an observation model for dependence,
705 source mixing and reference choice. Public mouse calcium imaging provides
706 thousands of spatially located neurons across waking, sleep and anesthe-
707 sia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-
708 estimation band used here. Site count, state coverage and bandwidth are
709 separate requirements.

710 **8 The cortical hypothesis and an awakening** 711 **test**

712 The cortical proposal identifies a physical source of coupling and asks whether
713 its changes help explain recovery. The tests in Section 7 determine what

714 observations must distinguish before a phase trace can support this identifi-
715 cation.

716 8.1 Physical identification and calibration

717 The cortical hypothesis identifies part of effective coupling with endogenous
718 extracellular electromagnetic interactions. Fields can modulate spike timing
719 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
720 propose coordination of distributed neural information (Pinotsis & Miller,
721 2022; McFadden, 2020). Phase organization under propofol motivates study-
722 ing this regime (Bastos et al., 2021), but does not measure coupling.

723 The formal field predicate requires a jointly continuous kernel, a normal-
724 ized substrate, coupling equal to the kernel average, and no single site pro-
725 viding more than half that strength. Toy constructions satisfy it. A cortical
726 realization must specify the scale at which the kernel and its reduction ap-
727 ply: calcium imaging reports different state-dependent modularity patterns
728 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

729 The spatial reduction is also testable. A row-sum aggregation at zero
730 diagonal tracks the uniform-coupling control over the tested spatial decay
731 range, while frequency heterogeneity displaces the operational threshold. The
732 fit window biases both estimates relative to $2D$; agreement with the control
733 is therefore the supported conclusion. Supplemental Material reports the
734 reduction, its numerical error and the distinct assumptions needed for the
735 kernel-average identification in Eq. (5).

736 The field commitment requires physical evidence beyond a choice of math-
737 ematical representation, and the available support comparisons do not supply
738 it. A finite matrix spread over positive-mass cells preserves its total cou-
739 pling and phase correlation exactly, while point-supported vanishing under
740 an atomless measure and a disadvantage from excluding a better-correlated
741 pair concern specified support constraints; Section 9.3 reads the same com-
742 parisons against a digital candidate. Neither supplies energy prices.

743 A spike-time shift alone supplies no coupling rate. The dimensionless
744 product of population size N , timing sensitivity s , field amplitude E and
745 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (15)$$

746 This rate describes how induced timing shifts accumulate into drift. Nei-
747 ther it nor the installation factor κ follows from a heat bound. Testing the
748 physical threshold requires independent estimates of the interaction kernel,
749 its aggregation and diffusion in the same cortical regime.

750 8.2 Extracellular geometry and recovery

751 The proposed control variable is extracellular geometry. Extracellular vol-
752 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
753 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
754 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively

755 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
 756 an effective-medium description predicts that a larger extracellular fraction
 757 raises conductivity and reduces field amplitude.

758 The coupling-gated hypothesis consequently predicts that extracellular
 759 contraction on awakening raises E and, holding the other factors in Eq. (15)
 760 fixed, raises effective coupling. This argument supplies a direction; the
 761 geometry–coupling relation, its magnitude and its role in human recovery
 762 require measurement.

763 For a smooth crossing at nonzero speed, evaluating the stationary branch
 764 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

765 This follows from Eq. (3) at fixed D and concerns the stationary branch
 766 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
 767 dently justified tracking approximation. The crossing controls in Section 7.2
 768 leave that approximation unsupported at onset, even though a separation of
 769 geometry and phase time scales can support it away from threshold. The
 770 awakening hypothesis therefore needs a calibrated dynamical prediction for
 771 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
 772 does not supply a recovery latency.

773 8.3 A protocol for the awakening experiment

774 The physical test has six requirements:

- 775 1. **Coupling and diffusion in the same units.** Estimate them from re-
776 solved phase dynamics, separating heterogeneity and observation noise.
777 Measure the drive or perturb the system to distinguish endogenous cou-
778 pling.
- 779 2. **Geometry calibrated to coupling.** Validate the geometry observ-
780 able and its relation to effective coupling, with uncertainty. The con-
781 ductivity argument supplies only the expected direction.
- 782 3. **Calibrated phase measurements.** Estimate r by resultant length
783 and a by unweighted regression of log density on centered cosine.
784 Choose site count and binning from a dependence-aware calibration
785 over the intended concentration range. The tested designs require
786 thousands of sites at the observed concentrations; temporal pooling
787 changes the spatial estimand.
- 788 4. **An independent recovery endpoint.** Measure behavioral recovery
789 separately from phase organization.
- 790 5. **A predeclared dynamical onset criterion.** Set a coherence thresh-
791 old above the finite-size floor and evaluate its latency under the mea-
792 sured coupling trajectory, including ramp speed or fluctuations. Record

793 a spatial observable beside the resultant, since a winding field reports
794 a low resultant at coupling that sustains local order (Section 7.3).

795 **6. Matched model comparison.** Compare coupling increase, diffusion
796 change, common drive and alternative recovery models under the same
797 observation and filtering assumptions.

798 Failure of the concentration–coherence relation over an informative range
799 constrains the density model. Failure of the calibrated geometry–coupling
800 association constrains the physical gate. Better prediction of independent
801 recovery by a competing mechanism counts against its proposed role. A
802 favorable outcome still leaves the content of coordinated activity to be ex-
803 plained.

804 **9 Discussion**

805 Two things have been established here, and they are of different kinds. The
806 constraints of Section 2 hold outright: synchrony does not deliver compatible
807 content, the resultant does not determine the state, a unique fixed point is
808 not a reconstruction, a region that cannot resolve a family cannot encode it,
809 and agreement about a change takes communication. Any account that reads
810 unity off phase coherence has to meet them, this one included. The account
811 set out in Section 1 is conditional at every step by contrast: each connection
812 in it is a hypothesis supplied rather than a result derived, and Table 1 names
813 all eight. Four of the ingredients assembled there are not standard — the

814 installed-energy condition, which makes a resource requirement necessary for
815 coherence on the stationary branch; the patch-local form of the coherence-
816 to-content bound, which is what keeps it from being vacuous; the stability
817 criterion and amplitude condition that settle when a low resultant reports a
818 winding field rather than weak coupling; and a machine-checked composition
819 carrying a witness that discharges all eight hypotheses at once, which is what
820 makes their conjunction non-empty rather than a list no object satisfies.
821 What follows takes the conditions in turn, as commitments the proposal
822 incurs, as measurements that would bear on them, and as questions the
823 framework leaves open.

824 **9.1 Unity and a minimal self as empirical commit-** 825 **ments**

826 We propose the state glued from compatible local descriptions as a corre-
827 late of conscious unity, and its accurate internal reconstruction as a correlate
828 of a minimal self. The motivations are that different contents belong to
829 one experienced situation and that the situation is experienced from a per-
830 spective. The conditional results identify relations a realization would need:
831 sufficient installed resources for the specified phase model, agreement about
832 shared content, and an encoding that supports reconstruction at the required
833 fidelity. Discharging E12–E89 for cortex would make the composition a state-
834 ment about cortex: that a cortical field above threshold relaxes to a cover

835 whose compatible local descriptions glue, and that the glued state is the
836 unique fixed point of a reconstruction map realized there. It would leave the
837 two identifications above exactly where they are, because neither is among
838 the eight and neither follows from their conjunction; they are the commit-
839 ment this subsection names, and Section 9.2 lists the three observations that
840 would count against it.

841 Compatibility allows different descriptions to contribute different infor-
842 mation. It also permits uncertainty and conflicting judgments: apparent cup
843 position and judged cup position can be distinct quantities in a consistent
844 description of their discrepancy. Which quantities are shared must be de-
845 termined independently. Choosing them solely to secure agreement would
846 deprive the proposal of empirical force.

847 The minimal-self proposal concerns the perceiver’s position and condition
848 within the situation, such as the cup’s location relative to a body that can
849 reach it. The encoding could be implicit and need carry no autobiograph-
850 ical narrative. Its proposed necessity is philosophical: even an unreflective
851 episode has a perspective. Gluing by itself requires no self-encoding, and re-
852 constructability alone supplies no functional role. Altering the encoded hand
853 or viewpoint while holding the external scene fixed should change estimates
854 of the perceiver’s relation to the cup as the readout predicts. Whether that
855 represented perspective is experienced remains a further question.

856 **9.2 What a test of the full proposal must establish**

857 Both uniqueness results concern specified objects. They choose neither the
858 cover nor the encoding, and do not select between candidate subjects. In-
859 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
860 here selection requires independent evidence about neural organization. A
861 constant readout can contract and fix a predetermined state while discard-
862 ing its input. The family criterion in Section 5.2 tests the sensitivity that
863 this construction lacks, provided the relevant states and fidelity are declared
864 independently.

865 A test must separately measure coupling relative to diffusion, content
866 compatibility and self-reconstruction, alongside an independent endpoint for
867 experience. Meeting all three without a unified episode would challenge suffi-
868 ciency. A unified episode with incompatible descriptions would challenge the
869 unity identification; one lacking the specified self-encoding would challenge
870 the reflexive requirement. Awakening, dreaming and dissociative states offer
871 possible settings, but absence of report cannot by itself define absence of
872 experience.

873 The next physical requirement is a calibrated geometry–coupling relation
874 and resolved phase dynamics. The installation bound contributes a nec-
875 essary resource condition where its hardware representation applies; cortical
876 mode prices and field-mediated coupling require measurement. For the repre-
877 sentational connection, an informative experiment asks whether a controlled
878 change in coordination alters independently decoded content agreement, with

879 controls for sensory information and decoder quality. The constructed-data
880 estimator establishes feasibility for specified contents. Neural local descrip-
881 tions, shared overlap encoders and the self-encoding have yet to be identified.
882 The population-size dependence of the phase-to-content bound further limits
883 its quantitative use, in the form Section 4.2 reads off the sheet.

884 The separation results explain why these measurements must be inde-
885 pendent. A stationary distribution leaves several recovery mechanisms indis-
886 tinguishable. Equal phase order can accompany incompatible descriptions
887 or different task information. The tested plasticity rule preserves coherence
888 while losing its designated template. Each finding concerns the specified
889 model and observable; their role is to delimit the evidence needed for the
890 proposed connections.

891 **9.3 What the conditions constrain in a digital candi-** 892 **date**

893 The results of Section 2 hold for any candidate, and for a large language
894 model (LLM) running on graphics processing units (GPUs) three of them are
895 checkable rather than merely stateable. That is where this framework bears
896 on claims made for such systems. It does not exclude them, and no result
897 here could; what it does is fix what a claim of unity or self-representation
898 commits its maker to measuring.

899 **Codes, not reports.** Reconstruction across a family is a counting con-
900 straint before it is anything else. If two macrostates the candidate is said
901 to represent satisfy $E(s) = E(t)$ while $d(s, t) > 2\varepsilon$, Eq. (10) rules out re-
902 constructing both within ε , whatever the readout. A system’s reports are
903 readouts. Identical verbal reports across two conditions are therefore consis-
904 tent with identical internal codes and settle nothing about the states, while
905 the converse claim — that the system represents its own condition across a
906 declared family — asserts that its internal code separates that family, which
907 is a statement about activations under intervention and not about text. The
908 measurement follows from the statement: declare the family and the fidelity
909 first, then test whether the encoding region distinguishes its members.

910 **The region, not the readout.** Where the encoding is read as the state’s
911 restriction to a region of the machine, the demand falls on the region before
912 any readout exists. For a decoder-only transformer the causal mask makes
913 the network an instance of the update model of Section 5: the activation at
914 position t and layer ℓ reads positions at most t at layer $\ell - 1$ and nothing
915 else, so a region’s causal past is fixed by the architecture rather than chosen
916 by interpretation. Two interventions outside that past that demand reports
917 more than 2ε apart cannot both be answered within ε from that region.
918 A claim that a model reports on its own processing consequently carries an
919 obligation to say which positions and layers the report’s causal past contains,
920 and a report that appears to track an intervention outside it indicates a

921 confound or a different region, not a faster channel.

922 **Agreement about what.** Unity requires the candidate’s local descriptions
923 to agree on independently identified shared quantities (Section 4). Common
924 timing, a shared residual stream and fluent output supply none of that, for
925 the reason Section 2 gives: coordination bounds content disagreement only
926 through encoders already related on overlaps, and which quantities two parts
927 of a model describe in common is a decoding question carrying the failure
928 modes of Section 7. The cover is a choice, and choosing it to secure agreement
929 empties the claim.

930 **What is not excluded.** Digital representations can satisfy compatibility,
931 so finite hardware is compatible with the gluing construction and none of
932 the above is a no-go for machines. The coupling-gated route E56–E78 is a
933 different matter: it requires a calibrated physical coupling, its noise regime
934 and a connection from coordination to content agreement, and an attention
935 matrix or a shared clock supplies none of these identifications. Comput-
936 ing a field model likewise leaves its realization in the computing substrate
937 to be established. Those physical obligations fall on the cortical proposal
938 identically.

939 The installed-energy condition of Section 3.2 is restricted in the same
940 way. It constrains a macroscopic candidate only through κ , which is hard-
941 ware data supplied by mode profiles and prices, and Supplemental Material
942 reads the condition against a cortical metabolic budget to return the conver-

943 sion factor it would require rather than a verdict. The hardware comparisons
944 there have similarly restricted scope: fixed wiring can impose a correlation
945 disadvantage, and point-supported kernels vanish under an atomless contin-
946 uum measure, but neither comparison makes the representation conditions
947 unattainable on GPUs. A universal exclusion would need a necessary condi-
948 tion for consciousness and a proof that every relevant implementation violates
949 it, and the present composition supplies neither.

950 Whether experience requires a living substrate is the explicit subject of
951 biological naturalism (Seth, 2026a,b). On that view the properties ground-
952 ing experience are those of self-sustaining organisms, and simulating them
953 does not secure them. The conditions stated here name no biological mech-
954 anism, but computation described as such does not satisfy them either: a
955 candidate must exhibit a physically realized coupling in a measured noise
956 regime, agreement about independently identified shared quantities, and an
957 internal reconstruction meeting a declared fidelity. Whether a realization
958 of those conditions that is not alive would be accompanied by experience
959 is the point at issue, and these results do not settle it. A related position
960 locates the difference in the mode of computation rather than in life as such:
961 biological systems compute across scales that do not separate, and along-
962 side discrete operations carry continuous-valued dynamics belonging to their
963 physical medium (Milinkovic & Aru, 2026). The coupling condition stated
964 here is one measurable instance of that requirement, fixed to a physical field
965 in a declared noise regime, and a description of such a field does not satisfy

966 it.

967 The neuroscientific case against *human-like* consciousness in text-only
968 LLMs concerns embodied sensory interaction, thalamocortical organization
969 and biological regulation linking perception and action to survival (Aru et al.,
970 2023). Their absence supports skepticism, conditional on their causal rele-
971 vance to experience. Those mechanisms are not requirements of the abstract
972 gluing and reconstruction definitions, and workspace-like representations re-
973 ported in some LLMs (Gurnee et al., 2026) supply candidate organization to
974 investigate. What this framework adds to that investigation is an order for
975 the questions: declare the family of states and the fidelity, fix the encoding
976 region and read its causal past off the architecture, and only then ask what
977 the system reconstructs. A claim that survives those three steps is a claim
978 about the machine; one that does not survive them was a claim about its
979 output.

980 **9.4 Explanatory scope**

981 Integrated-information, global-workspace and predictive-processing accounts
982 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
983 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a
984 candidate physical interaction and conditional relations between local content
985 and self-encoding. Gluing and reconstruction permit multiple substrates;
986 the electromagnetic commitment enters through physical calibration. The
987 formalization checks implications under their hypotheses, and its witnesses

988 establish realizability on declared models. Two things stand outside that
989 result and are not supplied by it: the eight hypotheses are discharged for
990 toy components rather than for cortex, and the identification of the glued
991 state with unity and of its reconstruction with a minimal self is a further
992 commitment that the eight do not contain.

993 The explanatory question is whether the specified relations occur in neural
994 activity and matter for experience. In the cup example, this means establish-
995 ing how coordination helps maintain one represented situation and how the
996 perceiver’s place within it contributes to the episode. The proposal identi-
997 fies candidate correlates and tests of that identification. Why those relations
998 would be accompanied by experience remains unexplained.

999 **Data and code availability**

1000 The Lean development, simulation scripts, tests and saved numerical sum-
1001 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
1002 `cortical_coherence`. Publication simulation values are generated from
1003 saved summaries without rerunning production sweeps. The exploratory
1004 EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-
1005 processing and selection are described in Supplemental Material. The repos-
1006 itory contains the reproducible analysis and does not redistribute the raw
1007 recordings.

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1011 **Use of AI tools**

1012 AI assistance to this work was substantial and is not separable by task. Sev-
1013 eral coding assistants and chat interfaces were used in an interleaved way
1014 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
1015 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
1016 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
1017 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
1018 accessed with the Hermes CLI. Between them they contributed to the Lean
1019 development, the simulation and analysis code, and the drafting and revision
1020 of both documents. Because the tools were used in combination and in alter-
1021 nation, no section, proof, figure or result is attributable to any one of them,
1022 and none is claimed to be. The author is responsible for all content, including
1023 everything a tool produced. Machine checking establishes the stated Lean
1024 results under their hypotheses; it does not validate biological interpretations
1025 or replace scientific review of generated code and prose.

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